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DEPARTMENT OF MECHANICAL ENGINEERING

I / II Semester B.E. Degree

Elements of Mechanical Engineering [22ME110/22ME210]

Faculty Name: Karthik MS

QUESTION BANK

UNIT - 2

Heat Engines:

- (1) Define I C and E C engines. List out the **Classification** of **internal combustion engine**.
- (2) Define the following: (i) **piston** (ii) **cylinder** (iii) **piston rings** (iv) **connecting rod** (v) **crank and crankshaft** (vi) **flywheel**
- (3) Define the following: (i) **Bore** (ii) **Top Dead Center (TDC) or Cover end** (iii) **Stroke length** (iv) **Swept (Stroke) Volume** (v) **Bottom Dead Center (BDC) or Crank end** (vi) **Clearance Volume** (vii) **Compression ratio** (viii) **piston speed**
- (4) Explain with a **neat sketch**, **Working principle** of **four stroke (4-s) Petrol engine (SI engine)** (**constant volume cycle engine**) (**gasoline engine**) along with **P-v diagram**.
- (5) Explain with a **neat sketch**, **Working principle** of **four stroke (4-s) Diesel engine (CI engine)** (**constant pressure cycle engine**) along with **P-v diagram**.
- (6) Explain with a **neat sketch**, **working principle** of **two stroke (2-s) Petrol engine (SI engine)**.
- (6) Explain with a **neat sketch**, **working principle** of **two stroke (2-s) Diesel engine (CI engine)**.
- (7) **Distinguish** between **(4-s) Petrol engine** and **(4-s) Diesel engine**.
- (8) **Distinguish** between **(2-s) engine** and **(4-s) engine**.
- (9) Define **gas turbine** and **classification** of **gas turbine**.
- (10) Explain with a **neat sketch**, **working principle** of **open cycle gas turbine (OCGT)**.
- (11) Explain with a **neat sketch**, **working principle** of **closed cycle gas turbine (CCGT)**.
- (12) **Distinguish** between **open cycle gas turbine** and **closed cycle gas turbine**.

Water Turbines:

- (1) **Define turbine and classification of turbine.**
- (2) **Define water (hydraulic) turbine and classification of hydraulic turbines.**
- (3) **Explain with a neat sketch, working principle of Impulse hydraulic turbine (Pelton turbine).**
- (4) **Explain with a sketch, working principle of Reaction hydraulic turbine (Kaplan turbine).**
- (5) **Distinguish between Impulse water turbine and Reaction water turbine.**